

BARNAUL, Russian Federation

WMO: 298380

Lat: **53.364N**Lon: **83.539E**Elev: **837**StdP: **14.26**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-23.6	-34.8	0.8	-28.6	-29.2	1.1	-23.0	28.8	20.0	24.3	18.6	4.7	330	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	86.2	65.8	83.1	64.8	80.5	63.7	69.6	80.3	67.7	78.0	66.1	76.7	7.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	99.1	74.5	64.1	92.5	73.1	62.3	86.6	71.3	34.2	80.2	32.7	78.0	31.3	77.0	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	20.0	17.5	-34.5	92.6	7.4	3.7	-39.8	95.3	-44.1	97.5	-48.3	99.5	-53.7	102.2		
(5)			-34.6	73.9	7.4	3.4	-39.9	76.4	-44.2	78.4	-48.4	80.4	-53.7	82.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	1.8	7.4	22.0	40.8	54.6	64.7	67.6	63.7	52.2	39.3	21.2	8.5		
	DBStd	25.46	14.73	13.22	11.26	10.45	9.71	7.14	5.66	6.62	8.20	9.63	14.03	14.10		
	HDD50	6508	1494	1192	870	310	65	3	0	2	70	351	863	1288		
	HDD65	10460	1959	1612	1335	727	341	90	39	104	391	796	1313	1753		
	CDD50	1815	0	0	0	33	209	444	547	426	136	20	0	0		
	CDD65	294	0	0	0	1	20	82	121	63	7	0	0	0		
	CDH74	2820	0	0	0	18	295	760	1013	628	104	2	0	0		
	CDH80	713	0	0	0	3	71	180	267	167	25	0	0	0		
	WSAvg	7.9	7.6	7.7	8.6	8.9	8.7	7.0	6.1	6.4	7.3	8.5	9.1	8.8		
	PrecAvg	17.8	0.9	0.7	0.7	1.1	1.5	2.2	2.9	1.9	1.4	1.4	1.6	1.3		
(15) Precipitation	PrecMax	22.0	1.7	1.5	1.7	3.8	3.9	4.7	5.7	3.8	3.5	4.0	3.5	2.9		
	PrecMin	12.1	0.2	0.0	0.1	0.1	0.4	0.5	0.9	0.5	0.2	0.4	0.5	0.5		
	PrecStd	2.8	0.4	0.4	0.4	0.7	0.8	1.0	1.3	0.8	0.8	0.7	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.3	35.7	50.0	76.3	86.4	88.6	89.9	91.1	83.4	71.3	51.9	36.1		
		MCWB	32.6	32.3	40.9	54.6	60.7	66.1	67.9	67.8	60.3	52.9	43.0	32.6		
	2%	DB	30.5	32.0	43.3	68.7	80.9	84.5	85.9	84.3	76.7	64.3	46.6	33.2		
		MCWB	28.8	29.5	36.5	51.7	58.7	64.2	67.8	65.8	58.8	49.9	40.5	31.2		
	5%	DB	26.4	28.8	39.6	63.3	76.9	81.8	82.6	80.5	71.3	58.8	41.3	29.8		
		MCWB	24.8	26.4	34.7	48.8	56.9	63.9	66.6	63.9	56.2	47.3	37.0	27.9		
	10%	DB	22.1	25.7	36.7	57.5	72.0	78.6	79.9	77.0	66.5	53.7	37.3	26.6		
		MCWB	20.5	23.6	33.1	46.1	55.3	62.9	65.6	62.5	54.0	45.1	34.5	25.1		
	0.4%	WB	32.8	33.2	41.3	56.5	64.9	70.1	72.8	70.5	63.6	54.1	45.1	33.5		
		MCDB	34.3	34.9	48.7	71.3	80.6	81.8	84.8	82.4	76.7	65.5	48.7	34.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.8	29.6	37.7	52.7	61.2	67.5	70.3	67.7	59.8	50.8	40.5	31.1		
		MCDB	30.5	31.6	42.0	66.0	76.1	79.0	80.6	79.6	72.3	61.6	45.5	32.8		
	5%	WB	24.6	26.5	35.3	49.9	58.8	65.9	68.7	65.7	57.6	48.4	37.3	27.8		
		MCDB	26.2	28.8	38.9	61.8	72.4	77.4	78.3	77.7	69.1	57.6	41.0	29.3		
	10%	WB	20.6	23.6	33.0	46.9	56.6	64.3	67.1	63.9	55.3	45.7	34.4	25.1		
		MCDB	22.0	25.6	36.2	57.0	69.3	75.7	76.5	74.4	64.8	52.7	37.1	26.6		
	5% DB	MDBR	15.9	17.4	17.7	19.1	21.8	20.0	19.2	20.5	20.1	16.5	12.8	14.6		
		MCDBR	14.7	16.5	18.3	27.8	28.3	24.8	23.8	26.2	28.1	26.3	14.6	15.0		
		MCWBR	13.9	14.7	13.4	15.5	13.7	11.1	10.0	11.1	13.8	14.8	10.6	14.2		
		MCWBR	14.7	16.3	15.8	25.5	24.2	21.2	20.2	23.2	24.4	23.8	13.2	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.330	0.381	0.408	0.407	0.404	0.429	0.405	0.369	0.355	0.337	0.303		
		taud	2.022	1.959	1.905	2.110	2.206	2.295	2.224	2.272	2.320	2.237	2.041	1.960		
	Ebn at Noon		200	231	245	259	266	269	259	258	253	225	179	165		
		Edh at Noon	24	36	47	44	42	39	41	37	31	27	24	22		
(44) All-Sky Solar Radiation	RadAvg		348	683	1124	1468	1775	1980	1788	1530	1082	564	285	235		
	RadStd		29	49	105	83	170	140	102	105	107	78	43	36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.92	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page